PS 01 — The Silent Recall

Domain: Manufacturing
Company: Precision Auto Parts Pvt. Ltd., Pune
Scale: 2 production lines · 3 shifts/day · ~8,000 units/day · 6 suppliers · 3 OEM clients
Annual Impact: ₹1,85,000+ per recall event · reputational risk with OEM clients

Background

Precision Auto Parts is a Tier-2 auto component manufacturer supplying brake pads and clutch assemblies to Tata, Mahindra, and Bajaj. In March 2024, an OEM raised a critical defect complaint — 340 brake pads showed surface delamination. The QC team spent 4.5 days manually cross-referencing shift registers, supplier invoices, and dispatch logs to trace the affected batch. By the time they identified the root batch, 2 more shipments had already left the factory.

The root cause: the company has zero digital traceability. Batch records are written in physical registers. Dispatch is logged in Excel. There is no link — digital or otherwise — between an incoming raw material lot, the production batch it feeds, the machine and shift that processed it, the QC outcome, and the dispatch order it ended up in.

Problem Statement

Design an AI-powered batch traceability system that creates a real-time digital thread connecting:

- Incoming raw material lot number
- Production batch (machine ID + shift + operator)
- QC inspection result and defect type
- Finished goods inventory
- Customer dispatch order

The system must answer this question in under 30 seconds:

"For dispatch order #D-1847, which raw material lot was used, which machine processed it, which shift, and what was the QC reading?"

Live Demo Requirements

- Trace dispatch order #D-1847 end-to-end: return raw lot number, supplier, machine, shift, QC defect rate
- Simulate a contamination alert on lot LOT-2023-114 — list all dispatch orders at risk, their OEM customers, and shipment dates
- Demonstrate the operator data entry interface for a shop floor worker logging a new batch

Data Files Provided

File Name	~Rows	Key Fields	Noise / Gotcha	Demo Anchor
raw_materials_log.csv	~2,400	receipt_date, supplier_id, lot_number, quantity_kg, quality_grade	8% missing lot numbers; 4 inconsistent date formats mixed throughout	LOT-2023-114 is the defective lot — must trace to downstream batches
production_log.csv	~5,400	date, shift, machine_id, batch_id, input_lot_ref, units_produced	12% rows missing batch_id (paper-to-Excel gaps)	BATCH-2023-0500 to 0503 consume LOT-2023-114
qc_inspection.csv	~5,400	batch_id, pass_fail, defect_type, defect_rate_pct	6 inconsistent label spellings for 'surface_delamination'	Batches 0500–0503 all FAIL with delamination
dispatch_log.csv	~1,800	order_id, customer_id, batch_ref, vehicle_number	batch_ref sometimes contains multiple batches (comma-separated)	D-1847 links to BATCH-2023-0500
supplier_master.csv	6 rows	supplier_id, supplier_name, material_supplied, lead_time_days, approved_status	Clean reference file	S03 supplied the adhesive in LOT-2023-114
defect_complaints.csv	3 rows	complaint_id, oem_id, affected_order_ids, defect_description, financial_impact_inr	Historical OEM complaints — ground truth for validation	C001 maps exactly to the March 2024 recall scenario

Constraints

- No RFID or barcode hardware budget — solution must work via manual data entry or QR code stickers (₹0 per unit hardware cost)
- Shop floor operators have Class 10 education — UI must be usable in simple English or Marathi
- Must integrate with existing Excel-based dispatch system — no ERP migration
- Any trace query must return results in under 30 seconds
- Must handle retroactive linking — connecting old paper records that have no current digital link
- Internet connectivity on shop floor is unreliable — system must function offline with sync capability

Deliverables

- D1: Working traceability system — web app or local Python app with search interface
- D2: Data cleaning pipeline that links the 4 noisy CSVs with documented assumptions for missing batch refs
- D3: Live demo — trace D-1847 showing full chain from dispatch back to raw material lot

- D4: Live demo — contamination alert on LOT-2023-114 with full list of affected dispatch orders
- D5: Operator data entry interface — offline-capable, Marathi/English
- D6: Written brief — how this system scales to 10 production lines (1 page)

Judging Criteria

Weight	Focus Area	What Judges Look For
40%	Data linking logic	How are missing batch_refs handled? Does the team use date+lot proximity, probabilistic matching, or ML? A simple inner join scores low. Creative, defensible imputation scores high.
30%	Trace speed & accuracy	Does the demo actually return the correct chain in <30 seconds? Are all linked records shown? Does the contamination alert catch all 4 affected dispatch orders?
30%	Operator UI usability	Would a Class 10 shop floor worker actually use this? Language, error handling, offline support, and simplicity are all evaluated.